Dmart (Mumbai, India) Warehouse Automation

Engineering Project Management (EMGT 5220)
Spring 2025 Term Project
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Team 1: Taskmasters

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PROBLEMS

Dmart's current warehouse operations in Mumbai rely heavily on manual processes, leading to:

- **Slow Order Fulfillment:** Manual picking and sorting and delay in product dispatch and delivery
- **Inventory Inaccuracy:** Lack of real-time tracking leads to stock mismatches and errors
- **High Operational Costs:** Significant labor requirements increase expenses and reduce efficiency
- **Inefficient Space Utilization:** Manual stacking and retrieval waste valuable warehouse space
- **Limited Scalability :**Current system cannot keep up with increasing demand and business growth

Purpose

Transform Dmart’s Warehouse Operations by:

- Integrating Robotics-Driven Automation
- Boosting Efficiency & Accuracy
- Reducing Manual Errors
- Enabling Scalable Inventory Management
- Supporting Long-Term Business Growth

GOALS

Deploy AMRs and AS/RS to streamline inventory handling

Frees Up Human Resources

Reduces Operational Costs

Enhance order fulfillment speed, inventory accuracy & space utilization

Objective

Automate Warehouse Operations

- Reduce human involvement in repetitive tasks using robotics
- Improve inventory accuracy
- Enable real-time data tracking and minimize errors
- Reduce labor costs and lower operational expenses through automation
- Speed up order fulfillment and streamline the sorting, picking and dispatching of goods using AMRs
- Optimize space utilization by using vertical and horizontal space effectively via AS/RS
- Ensure Scalable Infrastructure Design for future expansion and technology upgrades

Benefits to client

Improved Accuracy

- Real-time inventory tracking with RFID & WMS
- Reduced discrepancies, overstocking & stockouts
- Predictive restocking boosts availability

Workforce Productivity

- Staff can focus on strategic tasks over repetitive labor
- Training, change management, and support structure in place

Cost Efficiency

- Fulfillment costs lowered by 25%
- Space optimized by 30% via vertical storage (AS/RS)
- Scalable infrastructure for future expansions

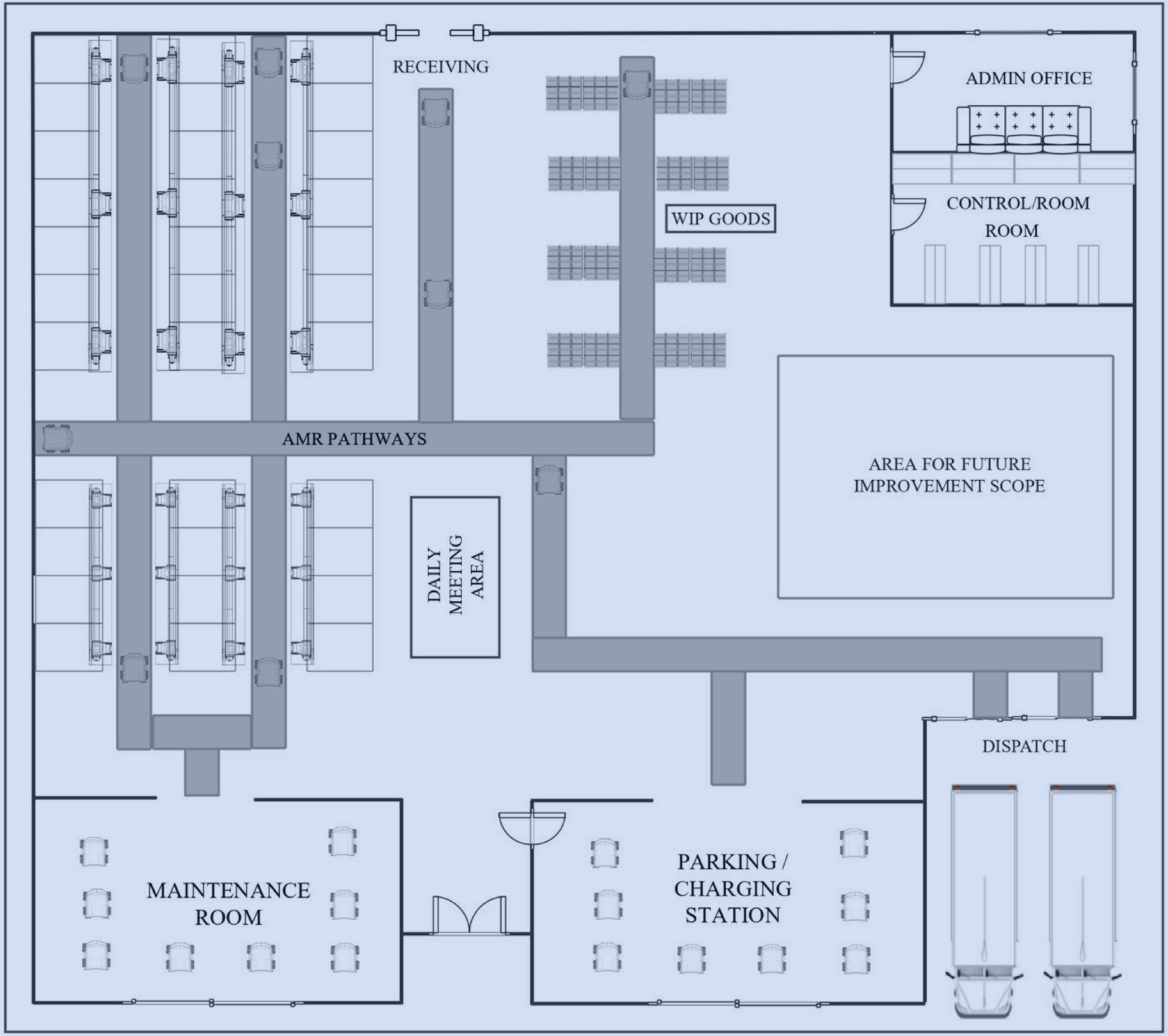
Business Impact

- Faster replenishment cycles ➤ Higher customer satisfaction
- Competitive edge in India’s dynamic retail market

How we will solve the problem

- Utilized **Microsoft Project** for project scheduling, tracking, and documentation
- Simulated layouts and flows using **CAD** to minimize bottlenecks
- Studied high-performing **automation systems** to adopt best practices and avoid common pitfalls
- Aligned automation design with **robotics** and **WMS** for seamless execution
- Designed **flexible systems** that adapt to future warehouse growth

WAREHOUSE FLOOR PLAN



APPROACH

Location: Maharashtra, India | 📐 Area: 15,000 m^2

01 Structure & Design

- Steel frame with 5.5m clearance
- M30 concrete floors, insulated roofing
- High-bay racking layout

02 Automation Systems

- 50 AMRs for internal transport
- 10 AS/RS for vertical inventory stacking
- Integrated conveyors & packing zone

03 Tech Infrastructure

- Central WMS with real-time tracking
- IoT sensors, industrial Wi-Fi
- Power backup (DG + UPS)

04 Compliance

- NBC & OSHA standards
- Fire safety, LED lighting, energy sensors

TECHNICAL OVERVIEW

CIVIL WORK

01 Site Preparation & Foundation

- Dismantling redundant structures, debris removal
- Soil excavation & backfill
- Isolated footing, reinforcement, shuttering & concrete work
- Leveling, compaction & warehouse flooring

02 Superstructure & Storage Infrastructure

- Roof and sidewall extensions for vertical storage
- Brick/blockwork for Control/Server and Maintenance Rooms
- Tiling, epoxy coating, floor screeding, and polishing

03 Structural Upgrades

- High-rise racking system installation
- Dedicated AMR pathways & floor markings
- Construction of robot parking and charging stations

04 Integration Readiness

- Reinforced floor to bear AMR and AS/RS loads
- Optimized layout for 2m aisles & 250 storage racks
- Installation of 10 AS/RS units with 8m high racking

05 Supporting Infrastructure

- Central control room for robotic & WMS monitoring
- Power backup installation (transformers, DG sets)
- Climate control (HVAC) for operational consistency

06 Workflow Enablement

- Dock redesign for smoother inbound/outbound logistics
- Pathways and zones allocated for picking, packing & dispatch
- Compliance with BMC, NFPA and DISH safety norms

TECHNICAL OVERVIEW

Mechanical and electrical

01 Implemented Autonomous Mobile Robots (AMRs) and Automated Storage and Retrieval Systems (AS/RS) for optimizing stock movement and inventory accuracy.

02 Detailed robot pathway planning within the warehouse layout to minimize congestion and ensure efficient material transport.

03 Installed a high-capacity electrical system with transformers, switchgear, and backup generators to support automation systems and critical operations.

04 Designed and implemented an efficient electrical layout for control rooms, charging stations, and automated systems.

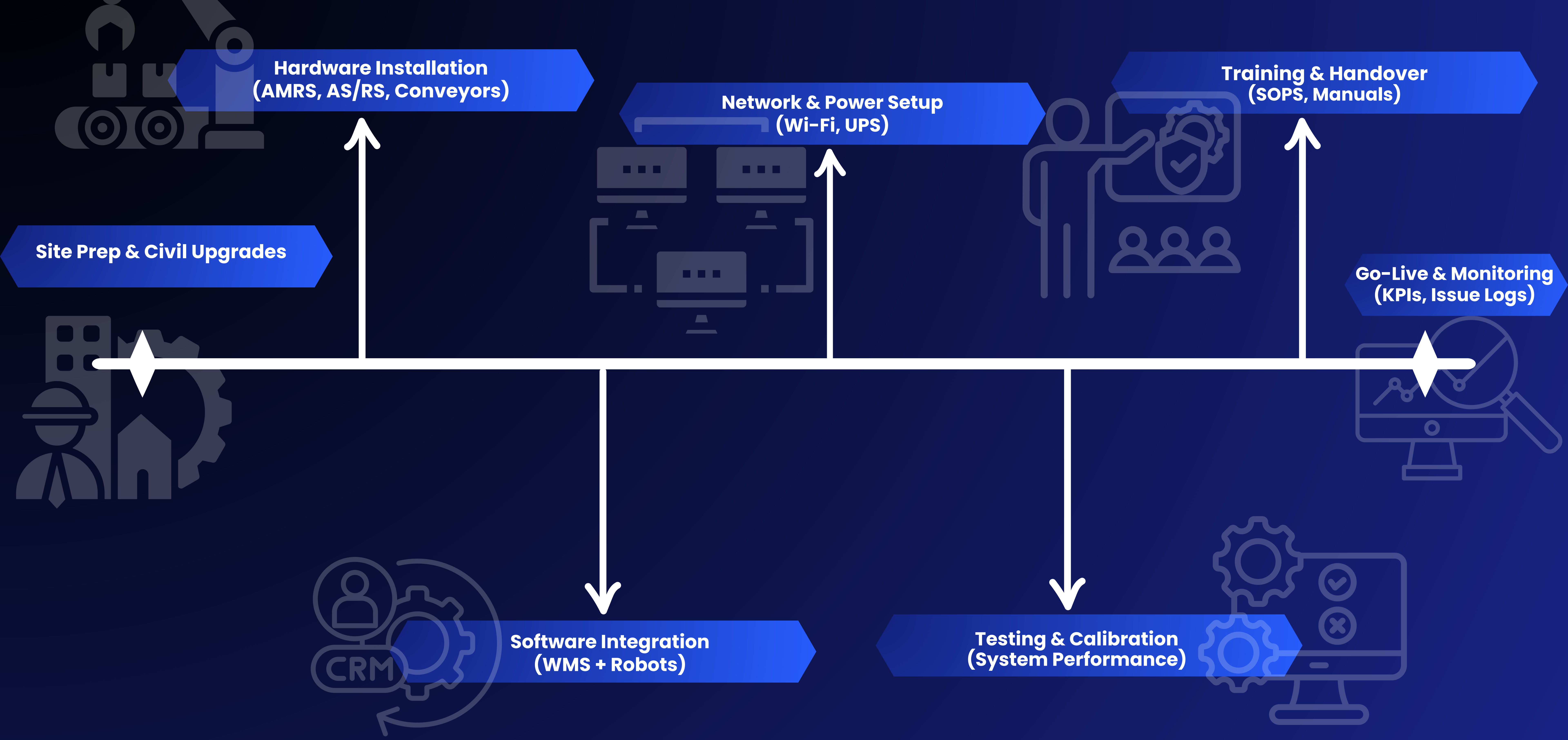
05 Integrated electrical infrastructure for automated systems such as AMR, AS/RS, and charging stations.

06 Ensured accurate alignment with Warehouse Management System (WMS) for seamless integration and real-time tracking.

07 Conducted full-scale testing and analysis of the robotic systems, refining the process based on performance data and system feedback.

08 Ensured compliance with safety regulations by implementing earthling systems, surge protection, and electrical safety protocols.

Technical Workflow Drart Warehouse Automation



IMPLEMENTATION PLAN

PHASE 1

- Assemble project team and define roles
- Set up communication plan and stakeholder engagement
- Draft project charter with scope, objectives, and KPIs

PHASE 2

- Conduct feasibility studies (architectural, structural, mechanical)
- Design warehouse layouts and automation system integration
- Plan procurement, workforce needs, and risk assessment

PHASE 3

- Source construction materials and automation hardware
- Procure electrical components and other essential equipment
- Select and negotiate contracts with vendors

IMPLEMENTATION PLAN

PHASE 4

- Civil works (foundation, superstructure, parking, pathways)
- Mechanical system installation (robots, conveyors, calibration)
- Electrical setup (transformers, wiring, power integration)

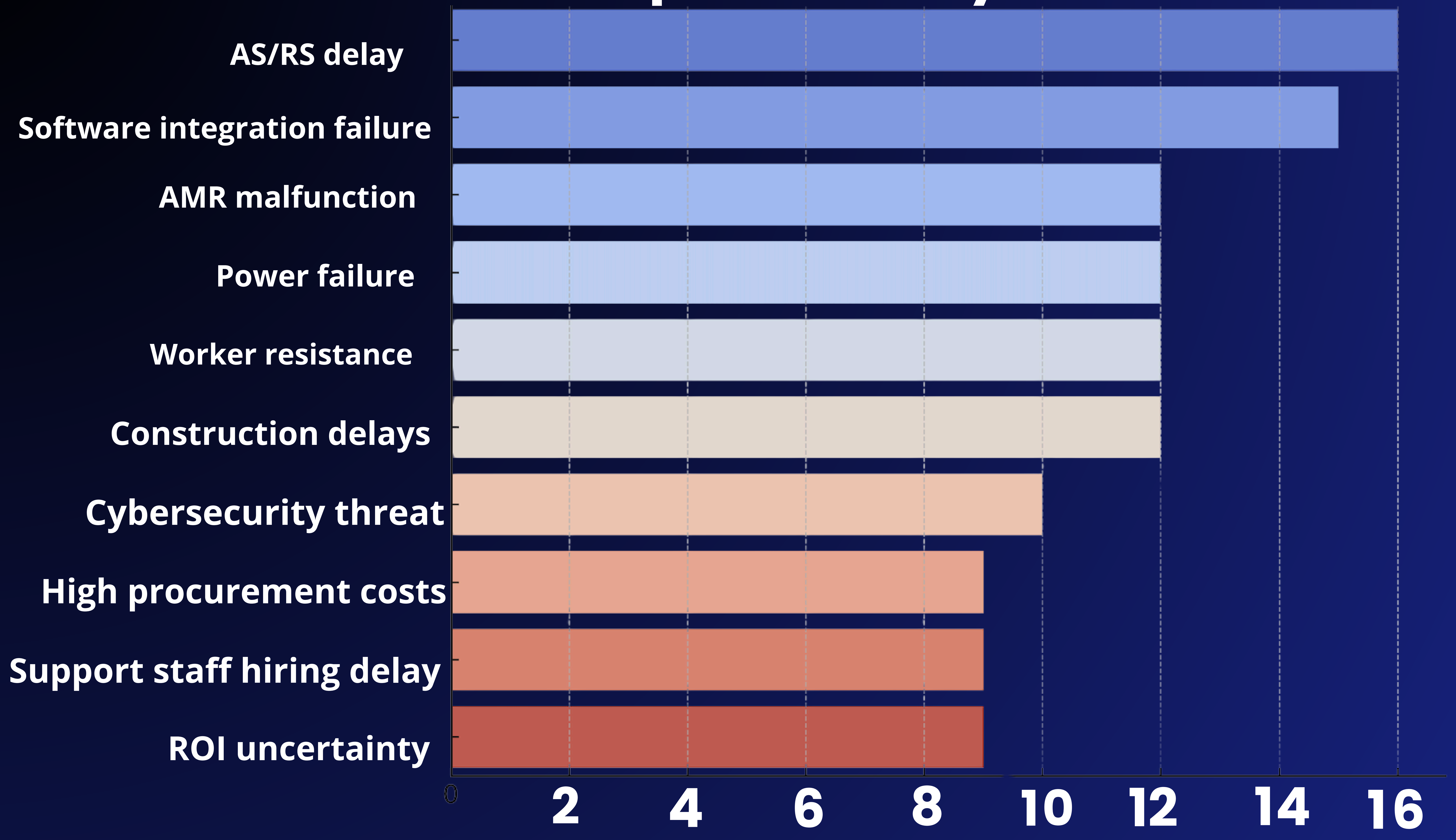
PHASE 5

- Conduct unit and system testing for automation performance
- Analyze KPIs, optimize automation paths, and refine systems
- Train end-users and collect feedback for system refinements

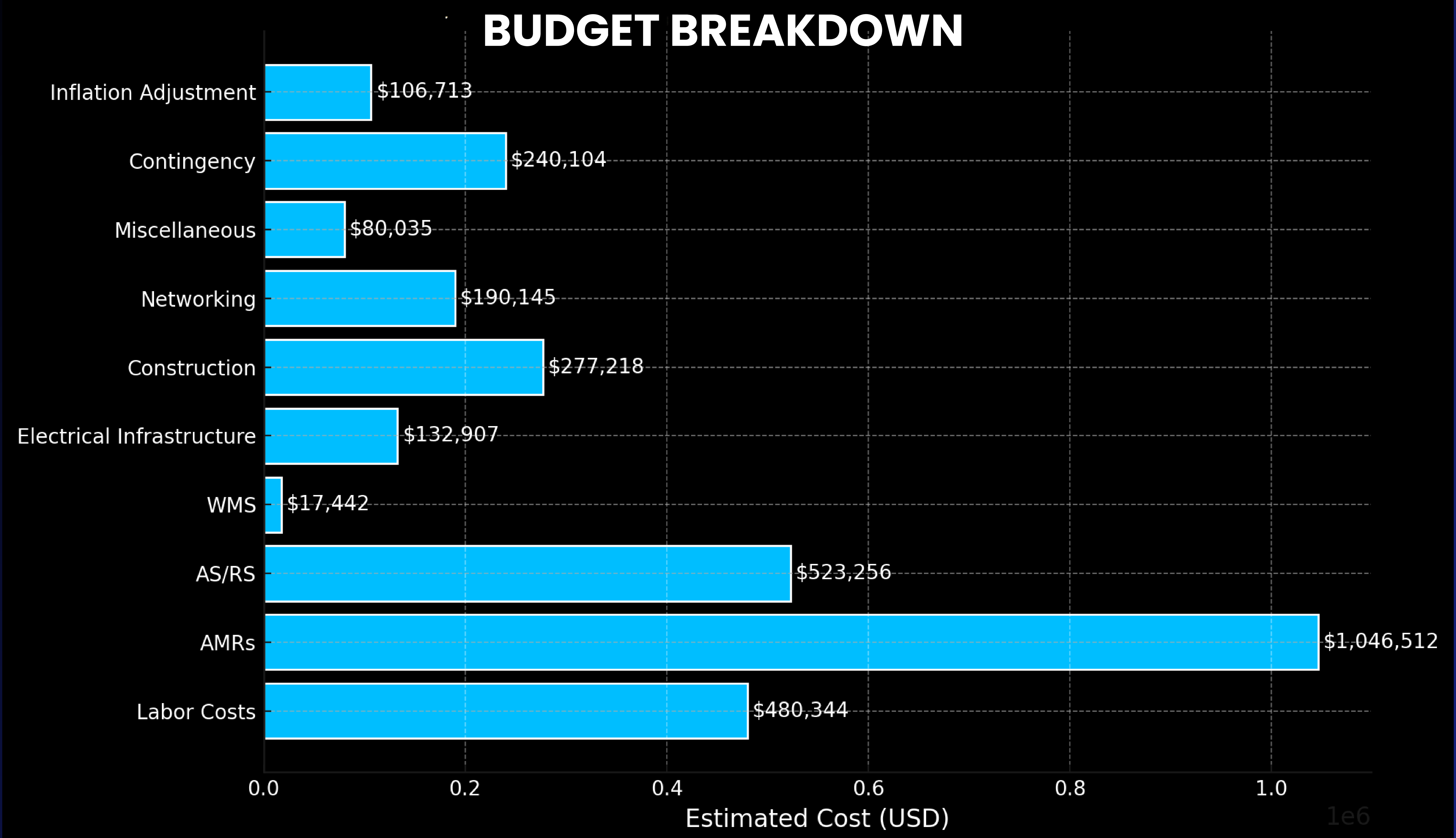
PHASE 6

- Full-scale system deployment and integration with WMS
- Staff training and change management for smooth transition
- Ongoing performance monitoring, scheduled maintenance, and vendor coordination

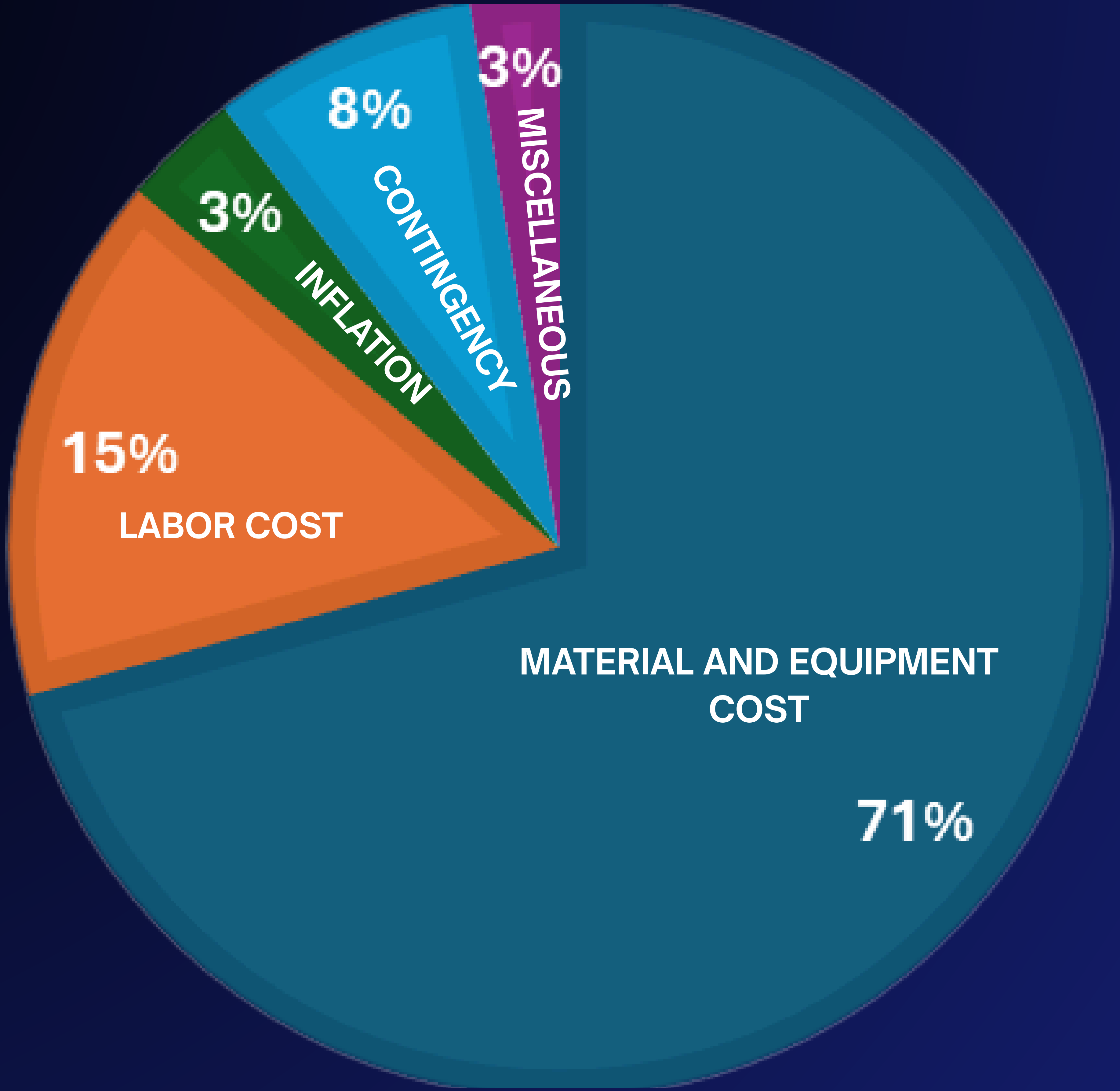
Top 10 Risks by Score



FINANCIAL PLAN



COST ALLOCATION



PROJECT EXECUTION

MONITORING

Progress Tracking
Schedule & Cost Control
Quality Assurance
Risk Management
Stakeholder
Communication
Change Management

CONTROLS

Schedule & Progress Control
Resource Control
Budget Control
Change Control
Risk Management
Issue Management

AUDIT

Performance Audit
Quality & Installation Audit
Procurement Audit
Change Audit
Risk Audit
Cybersecurity Audit
Post-Implementation Audit
Final Audit & Handover

CLOSURE

System Testing
Compliance
Documentation
Review
Site Cleanup
Handover

Dmart (Mumbai, India) Warehouse Automation

Problems: Manual operations cause delays, high costs, and poor scalability.

Goal

Automation using AMRs, AS/RS, and WMS for speed, accuracy, and growth

Benefits

- Cost reduction
- Space optimization
- Real-time tracking
- Skilled workforce

Impact

Boosts efficiency, supports expansion, and sets a model for future warehouses.

Thank You

For Your Attention

If you have any questions or
would like to discuss